ל4 — The Helmert-Simplicial Residue Ladder

Leonardo Murillo Montero

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Abstract

Situation. The trilogy *Identifying 3* gave three layers of the same object. ל1 supplied the native simplicial chart: keep the redundant coordinate, work in the mean-zero space, and read probability through the square-root sphere. ל2 supplied the generating functional: the categorical log-partition $\Phi = \log Z$, whose derivatives generate position, metric, cubic vertex, and the closure tower. ל3 supplied the arithmetic substrate: when the outcome labels carry $\mathbb{Z}/N$-structure, the character basis decomposes the mean-zero space into conductor packets and the cubic vertex obeys the exact selection rule

$$a + b + c \equiv 0 \pmod{N}.$$

Surprise. After the trilogy, the dimensional ladder itself reveals another hidden structure. If one descends an N-slot simplicial ledger by arithmetic mean removal, then climbs back by appending the balancing coordinate, the process is lossless. The resulting coordinates are the unnormalized Helmert balances, with k leading ones followed by the balancing entry $-k$:

$$h_k = (1, \dots, 1, -k, 0, \dots, 0), \qquad 1 \le k \le N-1.$$

This is the metric trick in ladder form: keep the integer incidence skeleton, and store length in the metric instead of leaking square roots into the coordinates.

Action. We formalize the reversible ladder, identify it with the unnormalized Helmert system, then deliberately remove the metric trick by orthonormalizing the ladder. The normalizers $\sqrt{k(k+1)}$ generate a multiquadratic field

$$K_N = \mathbb{Q}(\sqrt{\mathrm{sf}(k(k+1))} : 1 \le k < N),$$

where sf denotes squarefree kernel.

Result. The field generated by the normalized ladder up to rung N is exactly

$$K_N = \mathbb{Q}(\sqrt{p} : p \le N,\ p \text{ prime}).$$

Hence the field grows precisely on prime rungs:

$$K_N \neq K_{N-1} \quad \Longleftrightarrow \quad N \text{ is prime.}$$

The statement is structural: primes are the rungs at which orthonormal measurement of the simplicial ladder forces a new quadratic square-root generator. The metric trick hides this arithmetic in the metric; unit normalization exposes it as radical residue.

Harvest. ל4 therefore opens a fourth door after *Identifying 3*: the same mean-zero space that supports Fisher-Rao geometry, cumulant tensors, and conductor packets also supports a reversible balance ladder whose metric normalization externalizes number-theoretic square classes. The positive-cone version is the geometric-mean gauge, with multiplicative gauge removal corresponding to additive gauge removal under log. This connects simplicial coordinates, Aitchison log-ratio geometry, Fisher-Rao information geometry, and the squarefree arithmetic of prime-generated quadratic fields.

Keywords: simplicial coordinates, quadrays, Helmert coordinates, mean-zero space, metric trick, squarefree kernel, multiquadratic fields, primorial residues, geometric mean, Aitchison geometry, Fisher-Rao geometry, conductor packets, information geometry.

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0. The surprise after the trilogy

The trilogy was called *Identifying 3* because the number three appeared three times:

1. §1: for $N = 4$, the mean-zero simplicial space has dimension 3, so quadray coordinates carry ordinary 3D vector algebra.
2. §2: the third cumulant is the first information-geometric object beyond the Fisher-Rao metric.
3. §3: the cubic tensor is the first cumulant whose character-basis coefficients carry a modular selection rule.

§4 begins with another surprise of the same kind. The ladder that moves between N-slot simplicial ledgers and $(N-1)$-slot ledgers is not merely a pedagogical device. It is a coordinate system already known in statistics under the Helmert name, but the usual presentation normalizes it immediately and thereby hides the integer skeleton.

The simplicial view asks us not to normalize too early.

Keep the redundancy. Center by arithmetic mean removal. Append the balancing coordinate when climbing back. The reward is a lossless dimensional ladder with integer rungs.

Then, only afterward, ask what happens when Euclidean unit measurement is imposed.

The answer: primes appear as new square-root field generators.

That is the paper.

Notation

Symbol	Meaning
$\mathcal{L}_N = \mathbb{R}^N$	The N-slot simplicial ledger.
$\mathbf{1}_N$	The all-ones vector $(1, \dots, 1) \in \mathbb{R}^N$.
$\bar{q}$	Arithmetic mean $\bar{q} = \frac{1}{N}\sum_i q_i$.
H_N	Mean-zero $(N-1)$-space: $H_N = \{q \in \mathbb{R}^N : \sum_i q_i = 0\}$.
C_N	Centering map $C_N(q) = q - \bar{q}\mathbf{1}_N$.
m_k	Ladder packet coordinate at rung k.
$h_k^{(N)}$	Unnormalized Helmert ladder vector in $\mathbb{R}^N$.
$e_k^{(N)}$	Unit Helmert vector $h_k^{(N)}/\sqrt{k(k+1)}$.
J_N	The $N \times N$ all-ones matrix.
M_N	Exact-arithmetic metric $M_N = NI_N - J_N$.
$\mathrm{sf}(n)$	Squarefree kernel of n.
K_N	Quadratic field generated by normalized ladder constants up to rung N.

We deliberately write H_N, not H_0, when the ambient ledger size matters. Thus $H_N \subset \mathbb{R}^N$ has dimension $N-1$. This is the same object called H_0 in §1, §2, and §3, now indexed by ledger size.

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1. The reversible ladder

Let

$$q^{(N)} = (q_1, \dots, q_N) \in \mathcal{L}_N.$$

The descent step records the arithmetic mean

$$m_N = \frac{1}{N}\sum_{i=1}^{N} q_i,$$

centers the ledger

$$r^{(N)} = C_N(q^{(N)}) = q^{(N)} - m_N\mathbf{1}_N,$$

and rebalances downward by keeping the first $N-1$ centered entries:

$$q^{(N-1)} = (r_1^{(N)}, \dots, r_{N-1}^{(N)}).$$

No information has been lost, because the missing entry is determined by the mean-zero condition:

$$r_N^{(N)} = -\sum_{i=1}^{N-1} r_i^{(N)}.$$

Thus descent is reversible if the mean m_N is carried along.

Iterating downward gives a packet

$$\{m_1, m_2, \dots, m_N\}.$$

The bottom coordinate m_1 is the terminal one-slot ledger. It may be read as a mean or as a residual depending on orientation; both readings are useful. We keep the braces to mark the gauge-aware packet view.

The upward reconstruction is recursive:

$$q^{(1)} = (m_1),$$

and for $2 \le k \le N$,

$$r^{(k)} = (q^{(k-1)}, -\sum_{i=1}^{k-1} q_i^{(k-1)}),$$

$$q^{(k)} = r^{(k)} + m_k \mathbf{1}_k.$$

This is centering and rebalancing, all the way down and all the way back up.

Example 1.1

For

$$q^{(4)} = (2, 3, 4, 5),$$

descent gives

$$\{m_1, m_2, m_3, m_4\} = \left\{-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{7}{2}\right\}.$$

Climbing back:

$$q^{(1)} = \left(-\frac{1}{2}\right),$$

$$q^{(2)} = \left(-\frac{1}{2}, \frac{1}{2}\right) - \frac{1}{2}(1,1) = (-1,0),$$

$$q^{(3)} = (-1,0,1) - \frac{1}{2}(1,1,1) = \left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right),$$

$$q^{(4)} = \left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}\right) + \frac{7}{2}(1,1,1,1) = (2,3,4,5).$$

The ladder is lossless.

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2. The metric trick is the unnormalized Helmert ladder

Define, for $1 \leq k \leq N-1$,

$$h_k^{(N)} = (1, \ldots, 1, -k, 0, \ldots, 0) \in \mathbb{R}^N,$$

with k leading ones. Also define

$$h_N^{(N)} = \mathbf{1}_N.$$

These are the ladder vectors.

Proposition 2.1. Orthogonality

For $1 \leq j, k \leq N-1$,

$$h_j^{(N)} \cdot h_k^{(N)} = 0 \quad (j \neq k),$$

and

$$\|h_k^{(N)}\|^2 = k(k+1).$$

Also

$$h_k^{(N)} \cdot \mathbf{1}_N = 0.$$

Proof. Suppose $j < k$. The first j entries of both vectors contribute j. The $(j+1)$-st entry contributes $(-j)(1) = -j$, because $h_k^{(N)}$ still has value 1 there. All later entries contribute zero until the $(k+1)$-st entry, where $h_j^{(N)}$ is already zero. Hence the dot product is $j - j = 0$. The norm is

$$\|h_k^{(N)}\|^2 = k \cdot 1^2 + (-k)^2 = k + k^2 = k(k+1).$$

The sum of entries is $k - k = 0$, so $h_k^{(N)} \cdot \mathbf{1}_N = 0$. $\square$

Proposition 2.2. Direct reconstruction

Every $q \in \mathbb{R}^N$ has the expansion

$$q = m_N \mathbf{1}_N + \sum_{k=1}^{N-1} m_k h_k^{(N)},$$

where

$$m_N = \frac{1}{N} \sum_{i=1}^{N} q_i,$$

and

$$m_k = \frac{h_k^{(N)} \cdot q}{k(k+1)} = \frac{q_1 + \cdots + q_k - k q_{k+1}}{k(k+1)}.$$

Proof. The vectors $h_1^{(N)}, \ldots, h_{N-1}^{(N)}$ are mutually orthogonal and span the mean-zero space H_N, while $\mathbf{1}_N$ spans the shared-displacement line. Since they are mutually orthogonal and non-zero, the coefficient formulas are the usual orthogonal projection formulas. □

This is the clean algebraic heart of §4.

The metric trick says: do not divide by $\sqrt{k(k+1)}$ unless unit measurement is actually needed. Keep h_k integer, and carry the metric.

The exact-arithmetic metric from the quadray engineering guide generalizes as

$$M_N = N I_N - J_N.$$

On the mean-zero space H_N, this reduces to

$$u^T M_N v = N\, u \cdot v.$$

So in the ladder basis, the metric is diagonal:

$$h_j^T M_N h_k = N k(k+1) \delta_{jk}.$$

The integer skeleton is simple; the metric remembers the lengths.

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3. First divergence: the unit ladder and the quadratic field clock

Now deliberately remove the metric trick.

Define the unit Helmert vectors

$$e_k^{(N)} = \frac{h_k^{(N)}}{\sqrt{k(k+1)}}.$$

The square roots

$$\sqrt{k(k+1)}$$

are the normalization cost of forcing the integer ladder into Euclidean unit form.

Let

$$K_N = \mathbb{Q}(\sqrt{\mathrm{sf}(k(k+1))} : 1 \le k < N),$$

where $\mathrm{sf}(n)$ is the squarefree kernel of n.

Theorem 3.1. Prime rungs are field-extension rungs

For every $N \ge 2$,

$$K_N = \mathbb{Q}(\sqrt{p} : p \le N,\ p \text{ prime}).$$

Consequently,

$$K_N \neq K_{N-1} \quad \Longleftrightarrow \quad N \text{ is prime.}$$

More precisely, when N is prime,

$$[K_N : K_{N-1}] = 2,$$

and when N is composite,

$$K_N = K_{N-1}.$$

Proof. Every prime factor of $k(k+1)$ with $1 \le k < N$ is at most N, so

$$K_N \subseteq \mathbb{Q}(\sqrt{p} : p \le N,\ p \text{ prime}).$$

Conversely, let $p \le N$ be prime. For $p = 2$, the rung $k = 1$ contributes $\sqrt{1 \cdot 2} = \sqrt{2}$. For $p > 2$, take $k = p - 1$. The normalizer is

$$\sqrt{(p-1)p}.$$

All prime factors of $p-1$ are smaller than p, hence their square roots have already appeared by induction. Therefore $\sqrt{p}$ lies in K_N. This proves equality.

The second claim follows immediately: passing from $N-1$ to N adds $\sqrt{N}$ exactly when N is prime. The square classes of distinct primes are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so adjoining $\sqrt{N}$ for prime N doubles the degree. $\square$

This theorem should not be oversold as a prime algorithm. To decide field growth efficiently one still needs access to squarefree kernels, which is already factorization-shaped information. The theorem is structural:

primes are the rungs where unit measurement requires a new quadratic generator.

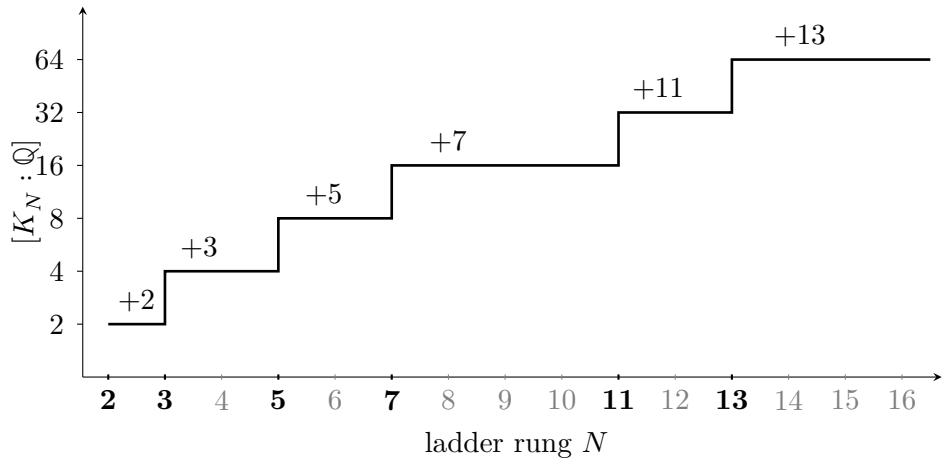


Figure 1: Prime-rung doubling of the multiquadratic radical field. The horizontal axis is the ladder rung N; the vertical axis is $[K_N : \mathbb{Q}] = 2^{\pi(N)}$ on a $\log_2$ scale. The step function doubles exactly at each prime N, where the unit normalizer $\sqrt{(N-1)N}$ adjoins the new generator $\sqrt{N}$ to K_{N-1}; at composite N the normalizer factors over primes already present and the field is unchanged. The labels $+2, +3, +5, +7, +11, +13$ mark each new quadratic generator $\sqrt{p}$. This is Theorem 3.1 made visible.

Examples

For $N = 4$:

$$K_4 = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

The rung 4 is composite, so no new generator appears:

$$\sqrt{3 \cdot 4} = \sqrt{12} = 2\sqrt{3}.$$

For $N = 5$:

$$\sqrt{4 \cdot 5} = 2\sqrt{5},$$

so

$$K_5 = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}).$$

A new prime rung has arrived.

For $N = 6$:

$$\sqrt{5 \cdot 6} = \sqrt{30} = \sqrt{2}\sqrt{3}\sqrt{5},$$

so the field does not grow.

For $N = 7$:

$$\sqrt{6 \cdot 7} = \sqrt{42} = \sqrt{2}\sqrt{3}\sqrt{7},$$

and $\sqrt{7}$ is new. The field grows again.

The radicals observed in explicit unit-frame computations,

$$\sqrt{7}, \sqrt{14}, \sqrt{21}, \sqrt{42}, \sqrt{70}, \sqrt{105}, \sqrt{210},$$

are therefore not accidents. They are visible elements of the new $\sqrt{7}$-coset inside the multiquadratic field generated by primes up to 7.

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4. Second divergence: multiplicative gauge and geometric-mean balances

The ladder has an additive form and a multiplicative form.

Let

$$x = (x_1, \dots, x_N) \in \mathbb{R}^N_{>0}.$$

The shared multiplicative magnitude is the geometric mean

$$g(x) = \left(\prod_{i=1}^{N} x_i \right)^{1/N}.$$

Multiplicative gauge removal is

$$x_i \mapsto \frac{x_i}{g(x)},$$

and it lands in the product-one space

$$\prod_{i=1}^{N} \frac{x_i}{g(x)} = 1.$$

Taking logs turns this into arithmetic mean removal. Let

$$y_i = \log x_i.$$

Then

$$\log \frac{x_i}{g(x)} = y_i - \frac{1}{N} \sum_{j=1}^{N} y_j.$$

Hence

$$\boxed{\text{multiplicative gauge removal} \quad \overset{\log}{\longleftrightarrow} \quad \text{additive gauge removal.}}$$

The Helmert ladder now becomes a ladder of geometric-mean comparisons. Compute

$$h_k \cdot y = \sum_{i=1}^{k} \log x_i - k \log x_{k+1} = \log \frac{x_1 x_2 \cdots x_k}{x_{k+1}^k}.$$

Equivalently,

$$h_k \cdot y = k \log \frac{\mathrm{GM}(x_1, \ldots, x_k)}{x_{k+1}}.$$

The unit version is

$$e_k \cdot y = \sqrt{\frac{k}{k+1}} \log \frac{\mathrm{GM}(x_1, \ldots, x_k)}{x_{k+1}}.$$

This is precisely the form of a sequential log-ratio balance: compare the geometric mean of the previous block against the next part.

Thus the same ladder has three readings:

additive ledger: arithmetic mean removal,

positive cone: geometric mean removal,

log space: Helmert/Aitchison balances.

This is the bridge to information geometry. §2's softmax is

$$p_i = \frac{e^{\eta_i}}{\sum_j e^{\eta_j}}.$$

The additive gauge

$$\eta \mapsto \eta + c\mathbf{1}$$

becomes multiplicative scaling of positive weights

$$e^{\eta_i} \mapsto e^c e^{\eta_i}.$$

Softmax removes that multiplicative gauge by sum-normalization. The geometric-mean gauge removes it by product-normalization. Both live on the same positive cone; they choose different sections.

§1 emphasized the square-root section

$$p_i \mapsto a_i = \sqrt{p_i},$$

which turns probability into sphere geometry. §4 adds the log-ratio section

$$x_i \mapsto \log x_i - \frac{1}{N}\sum_j \log x_j,$$

which turns positive composition into mean-zero balance geometry.

These are not rival geometries. They are two observer choices on positive measure.

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5. Third divergence: conductor packets and radical residues

§3 showed that when labels carry $\mathbb{Z}/N$-structure, the complexified mean-zero space decomposes into additive conductor packets

$$H_N \otimes \mathbb{C} = \bigoplus_{d|N,\ d>1} H_d.$$

The packet H_d has dimension

$$\dim_{\mathbb{C}} H_d = \varphi(d),$$

and the cubic vertex at the uniform obeys

$$T(\chi_a, \chi_b, \chi_c) = 1 \quad \Longleftrightarrow \quad a + b + c \equiv 0 \pmod{N}$$

for non-trivial characters.

That is the arithmetic of cyclic labels.

§4 finds a different arithmetic in the same mean-zero space: the arithmetic of metric normalization. The normalized ladder lives in the multiquadratic field

$$K_N = \mathbb{Q}(\sqrt{p} : p \le N,\ p \text{ prime}).$$

A natural basis for this field over $\mathbb{Q}$ is

$$\{\sqrt{d} : d \mid P_N,\ d \text{ squarefree}\},$$

where

$$P_N = \prod_{p \le N} p$$

is the primorial over primes up to N.

Thus the divisor lattice appears twice:

1. In §3, divisors $d \mid N$ classify character conductors of $\mathbb{Z}/N$.
2. In §4, squarefree divisors $d \mid P_N$ classify radical basis elements of the normalized ladder field.

These are not the same decomposition. One is Fourier-arithmetic inside a fixed cyclic group. The other is metric-arithmetic accumulated by climbing the dimensional ladder.

But they rhyme.

§3 says:

order 3 detects the additive ring structure of $\mathbb{Z}/N$.

§4 says:

unit measurement detects the prime square classes accumulated up to N.

The first is a selection rule. The second is a field-extension rule.

Together they suggest a larger arithmetic picture:

$$\text{mean-zero simplicial geometry} \quad \begin{cases} \text{Fourier basis} & \to \text{conductor packets,} \\ \text{Helmert ladder} & \to \text{quadratic radical packets.} \end{cases}$$

This is not yet a theorem connecting the two packet systems. It is a map of the terrain.

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6. Convergence of the ladder with the trilogy

The trilogy handed forward three objects:

Paper	Object	What it exposed
ל1	Mean-zero simplicial chart and square-root sphere	Fisher-Rao geometry as sphere geometry; quadray $N = 4$ as 3-space.
ל2	$\Phi = \log Z$	The cumulant tower: position, metric, cubic vertex, closure defects.
ל3	Character basis of $\mathbb{Z}/N$	Conductor packets and the cubic selection rule.

ל4 adds a fourth:

Paper	Object	What it exposes
ל4	Helmert-simplicial residue ladder	Lossless dimensional balance, geometric-mean gauge, and prime rungs as metric field extensions.

The unifying diagram is:

$$\mathbb{R}^N \xrightarrow{\text{arithmetic mean removal}} H_N \xrightarrow{\text{metric choice}} \begin{cases} \text{integer ladder} & \text{metric trick,} \\ \text{unit Helmert ladder} & \text{radical residues,} \\ \text{character basis} & \text{conductor packets,} \\ \text{square-root sphere} & \text{Fisher-Rao geometry.} \end{cases}$$

The positive-cone version is:

$$\mathbb{R}^N_{>0} \xrightarrow{\log} \mathbb{R}^N \xrightarrow{\text{arithmetic mean removal}} H_N.$$

Equivalently,

$$\mathbb{R}^N_{>0} \xrightarrow{\text{geometric mean removal}} \{x : \prod_i x_i = 1\} \xrightarrow{\log} H_N.$$

This is the language we were missing: additive gauge, multiplicative gauge, spherical probability, and arithmetic conductor structure all meet in the same mean-zero space.

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7. Theorems and forward program

The results above opened several doors. Some are no longer conjectural once the gauge-corridor and Fourier-Mellin viewpoint are made explicit. We separate the theorem-level content from the remaining forward problems.

Theorem 7.1. Gauge sections of the positive projective cone

Let

$$\mathbb{P}_+^{N-1} := \mathbb{R}_{>0}^N / \mathbb{R}_{>0}^\times$$

be the positive projective cone. The sum-one simplex section

$$\sigma_1([x]) = \frac{x}{\sum_i x_i}$$

and the product-one section

$$\sigma_0([x]) = \frac{x}{G(x)}, \qquad G(x) = \left(\prod_i x_i\right)^{1/N},$$

are well-defined global gauge sections of $\mathbb{P}_+^{N-1}$. More generally, the power-mean gauges

$$\sigma_r([x]) = \frac{x}{M_r(x)}$$

for $r \in \mathbb{R} \cup \{\pm\infty\}$, with $r = 0$ interpreted as the geometric mean, select one representative from each positive ray whenever the corresponding mean is finite.

Proof. If $x \sim \lambda x$ with $\lambda > 0$, then $\sum_i(\lambda x_i) = \lambda \sum_i x_i$ and $G(\lambda x) = \lambda G(x)$, so both displayed formulas are independent of the chosen representative. Each positive ray meets the sum-one section exactly once, at $x/\sum_i x_i$, and meets the product-one section exactly once, at $x/G(x)$. The same homogeneity argument applies to every power mean: $M_r(\lambda x) = \lambda M_r(x)$. □

Thus the square-root map of §1 and the log-ratio map of §4 are not rival geometries. They are two gauge choices on the same positive projective object.

Theorem 7.2. Helmert-Fourier coefficient factorization

Identify the outcome labels with $\mathbb{Z}/N = \{0, \dots, N-1\}$, let

$$\omega = e^{2\pi i/N}, \qquad \chi_c(s) = \omega^{cs},$$

and let h_k have k leading ones, entry $-k$ at slot k, and zeros afterward. Define the normalized Helmert-Fourier coefficient

$$U_{kc} := \frac{1}{\sqrt{N\,k(k+1)}} \sum_{s=0}^{N-1} h_k(s)\chi_c(s), \qquad 1 \le k \le N-1.$$

For $c \neq 0$,

$$U_{kc} = \frac{1}{\sqrt{N\,k(k+1)}} \left(\frac{1-\omega^{kc}}{1-\omega^c} - k\omega^{kc} \right).$$

The prefactor is the radical Helmert factor; the bracketed term is the cyclotomic conductor factor. For $c = 0$, $U_{k0} = 0$. Hence every transfer coefficient lies in the compositum

$$K_N\,\mathbb{Q}(\zeta_N),$$

where $K_N = \mathbb{Q}(\sqrt{p} : p \le N,\ p \text{ prime})$.

Proof. By definition,

$$\sum_s h_k(s)\chi_c(s) = \sum_{s=0}^{k-1} \omega^{cs} - k\omega^{kc}.$$

For $c \neq 0$, the first term is the geometric sum

$$\sum_{s=0}^{k-1} \omega^{cs} = \frac{1-\omega^{kc}}{1-\omega^c}.$$

For $c = 0$, the sum is $k - k = 0$. The normalization factor belongs to K_N: its squarefree part uses only primes dividing $Nk(k+1)$, all of which are at most N. The bracketed term belongs to $\mathbb{Q}(\zeta_N)$. □

This proves the entry-level conductor-residue dictionary: a Helmert-Fourier transfer coefficient is radical arithmetic from the dimensional ladder times cyclotomic arithmetic from the conductor basis.

Theorem 7.3. Cubic Helmert selection rule

At the uniform distribution, the categorical third cumulant contracted against mean-zero vectors is

$$T_u(v, w, z) = \frac{1}{N} \sum_{i=1}^{N} v_i w_i z_i.$$

For unnormalized Helmert vectors, sort the indices $a \le b \le c$. Then

$$T_u(h_a, h_b, h_c) = \begin{cases} \dfrac{a(a+1)}{N}, & a = b < c, \\ \dfrac{a - a^3}{N}, & a = b = c, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. If $a < b \leq c$, the first a entries contribute a, while the balancing entry of h_a contributes $-a$; all later entries vanish because h_a is zero there. Hence the total is 0. If $a = b < c$, the first a entries contribute a, and the balancing entry contributes $(-a)(-a)(1) = a^2$, giving $a(a+1)$. If $a = b = c$, the first a entries contribute a, and the balancing entry contributes $(-a)^3 = -a^3$. Multiplying by $1/N$ gives the displayed formula. □

This is the real Helmert-basis analogue of §3's modular cubic selection rule. The Fourier basis expresses the rule as character conservation; the Helmert basis expresses it as triangular balance incidence.

Remark 7.4. No algorithmic speed claim

The field-growth criterion

$$K_N \neq K_{N-1} \quad \Longleftrightarrow \quad N \text{ is prime}$$

is structural, not a faster prime sieve. Recognizing the new square class requires squarefree-kernel information, which is already factorization-shaped information. What the theorem supplies is a representation: prime rungs are the places where unit measurement forces a new quadratic generator.

Forward Problem 7.5. Fine radical support of canonical ladder lifts

Theorem 3.1 gives the invariant field K_N. A finer question remains: for a fixed recursive unit-frame convention and a chosen ladder vector, which square classes inside K_N actually appear?

For canonical first-axis ladder packets, examples suggest that the radical support appearing at a prime rung p lies in the new coset

$$\sqrt{p}\, K_{p-1} \subset K_p,$$

while composite rungs recombine previous square classes. This is observed in examples such as

$$N = 5: \quad \sqrt{5}, \sqrt{10}, \sqrt{30},$$

and

$$N = 7: \quad \sqrt{7}, \sqrt{14}, \sqrt{21}, \sqrt{42}, \sqrt{70}, \sqrt{105}, \sqrt{210}.$$

Because the exact support depends on the recursive convention and on the chosen ladder vector, this is a support-classification problem rather than another field-growth theorem.

Forward Problem 7.6. Packet-level conductor-residue dictionary

Theorem 7.2 proves factorization entry by entry. The remaining dictionary is packet-level: describe the full Helmert-Fourier transfer matrix as a structured object over conductor packets, ideally as a Gauss-sum / Vandermonde / discrete-Fourier hybrid with explicit block behavior.

This is where the two divisor lattices should meet:

1. divisors $d \mid N$ controlling additive conductor packets in §3;
2. squarefree divisors $d \mid P_N$ controlling radical basis elements in K_N.

The dictionary is not equality. It should relate two ways arithmetic enters H_N: through cyclic label structure at fixed N, and through metric normalization across the Helmert flag

$$H_2 \subset H_3 \subset \cdots \subset H_N.$$

Forward Problem 7.7. Higher cumulant balances in Helmert coordinates

Theorem 7.3 settles the cubic order. Higher categorical cumulants of §2, expressed in unnormalized Helmert coordinates, should admit closed triangular formulas whose coefficients separate:

1. integer incidence data from the ladder;
2. metric normalization data from the choice of unit basis.

From order 4 onward the connected cumulant formulas require partition-subtraction terms, so the higher-order Helmert selection rules should be stated for connected cumulants rather than raw powers.

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8. Worked example: the first ladder axis through seven slots

With the metric trick,

$$\{1,0,0,0,0,0,0\} \mapsto (1,-1,0,0,0,0,0).$$

All arithmetic remains in the metric

$$M_7 = 7I - J.$$

Without the metric trick, unit-frame coordinates live in

$$K_7 = \mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5},\sqrt{7}).$$

A fixed recursive unit-simplex convention produces terms including

$$\sqrt{7},\sqrt{14},\sqrt{21},\sqrt{42},\sqrt{70},\sqrt{105},\sqrt{210}.$$

The metric-trick expression is integer-simple. The unit expression is arithmetic-rich.

Both are the same geometry viewed through different measurement conventions.

This is the core interpretive lesson:

The metric trick keeps coordinates integral by moving arithmetic into the metric.

Unit normalization moves arithmetic out of the metric and into radical coordinates.

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9. Relation to known names

This paper deliberately uses the language of simplicial ledgers and gauge because the conceptual work began from quadrays and the dimensional ladder. Still, several standard mathematical objects are present.

1. The vectors h_k are the unnormalized Helmert contrast vectors.
2. The unit vectors e_k are the Helmert orthonormal basis.
3. The log-positive version is a sequential log-ratio balance from compositional data analysis.
4. The product-one section is the centered log-ratio gauge of Aitchison geometry.
5. The mean-zero space H_N is the real standard representation of the symmetric group S_N.
6. The integer mean-zero points form the root lattice A_{N-1}, though the Helmert ladder basis is an orthogonal rational basis rather than the usual simple-root basis.
7. The fields K_N are multiquadratic extensions with Galois group

$$\mathrm{Gal}(K_N/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^{\pi(N)}.$$

The novelty is not that these objects were unknown. The novelty is their alignment inside one simplicial ladder:

$$\begin{aligned}\text{quadray intuition} &\to \text{mean-zero balance} \to \text{metric trick} \\ &\to \text{prime field residues} \to \text{information geometry.}\end{aligned}$$

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10. Coda

ל1 said: do not delete the redundant coordinate.

ל2 said: one scalar functional generates the tower.

ל3 said: the cubic vertex is where arithmetic first speaks.

ל4 says: do not normalize the ladder too early.

If we keep the integer ladder and carry the metric, the simplicial system remains clean:

$$\{1, 0, \ldots, 0\} \mapsto (1, -1, 0, \ldots, 0).$$

If we insist on unit Euclidean measurement, the hidden arithmetic appears:

$$\sqrt{k(k+1)}, \qquad \mathbb{Q}(\sqrt{p} : p \le N,\ p \text{ prime}).$$

So the same object has two faces:

integer balance skeleton

and

quadratic prime-residue measurement field.

This is the post-trilogy surprise. The dimensional ladder is not only the path from simplicial N to Cartesian $N-1$. It is also a gauge machine, a log-ratio machine, and a prime-field clock.

The next work is to decide how much of this is merely beautiful language, and how much can be made into new mathematics.

ל4 opens that door.

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